

**Rino R. Mukti**
Division of Inorganic and Physical Chemistry
FMIPA ITB

Week 11

KI1101 General Chemistry 1

# Molecular Structure

# Lewis Symbols

- Electron bookkeeping method
- Way to keep track of $e^-$'s
- Write chemical symbol surrounded by dots for each $e^-$

| **Group #** | **IA** | **IIA** | **IIIA** | **IVA** |
|---|---|---|---|---|
| **Valence $e^-$'s** | **1** | **2** | **3** | **4** |
| **$e^-$ conf'n** | $ns^1$ | $ns^2$ | $ns^2np^1$ | $ns^2np^2$ |
| | H· | ·He· | | |
| | Li· | ·Be· | ·B· | ·C· |
| | Na· | ·Mg· | ·Al· | ·Si· |

# Lewis Symbols

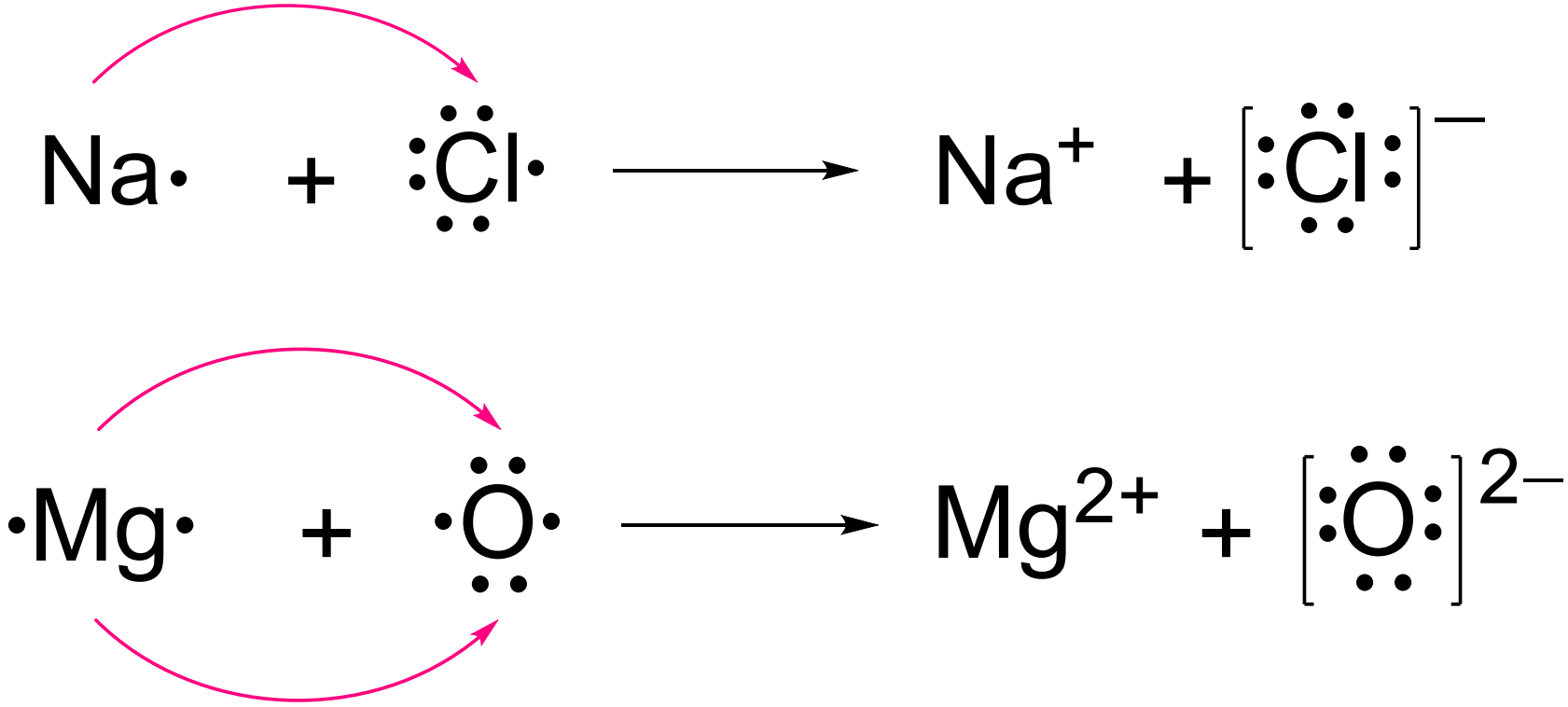
| Group # | VA | VIA | VIIA | VIIIA |
|---|---|---|---|---|
| Valence e⁻'s | 5 | 6 | 7 | 8 |
| e⁻ conf'n | $ns^2np^3$ | $ns^2np^4$ | $ns^2np^5$ | $ns^2np^6$ |
| | | | | ·He· |
| | ·N· | ·O· | :F· | :Ne: |
| | ·P· | ·S· | :Cl· | :Ar: |

For the representative elements

Group # = # valence e⁻'s

# Lewis Symbols

- Can use to diagram $e^-$ transfer in ionic bonding



# Covalent Compounds

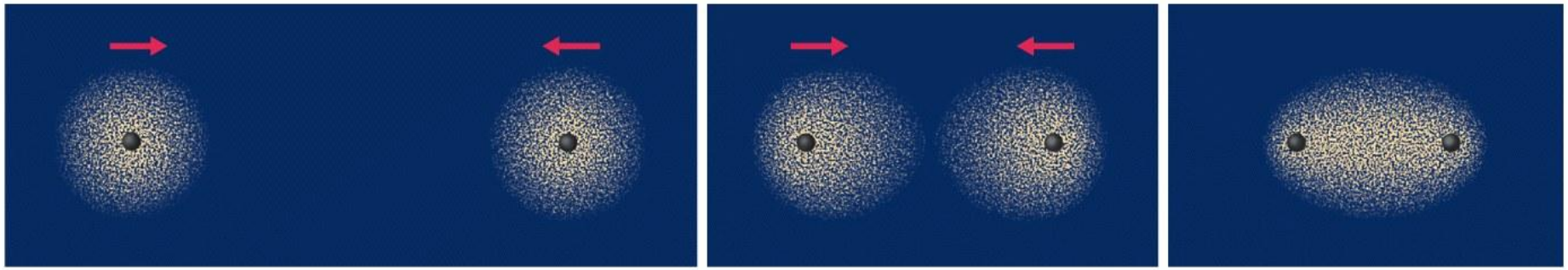
- Form individual separate molecules
  - Atoms bound by sharing $e^-$'s
- Do not conduct electricity
- Often low melting point

## Covalent Bonds

- Shared pairs of $e^-$'s between 2 atoms
- 2 H atoms come together, Why?

# Covalent Bond

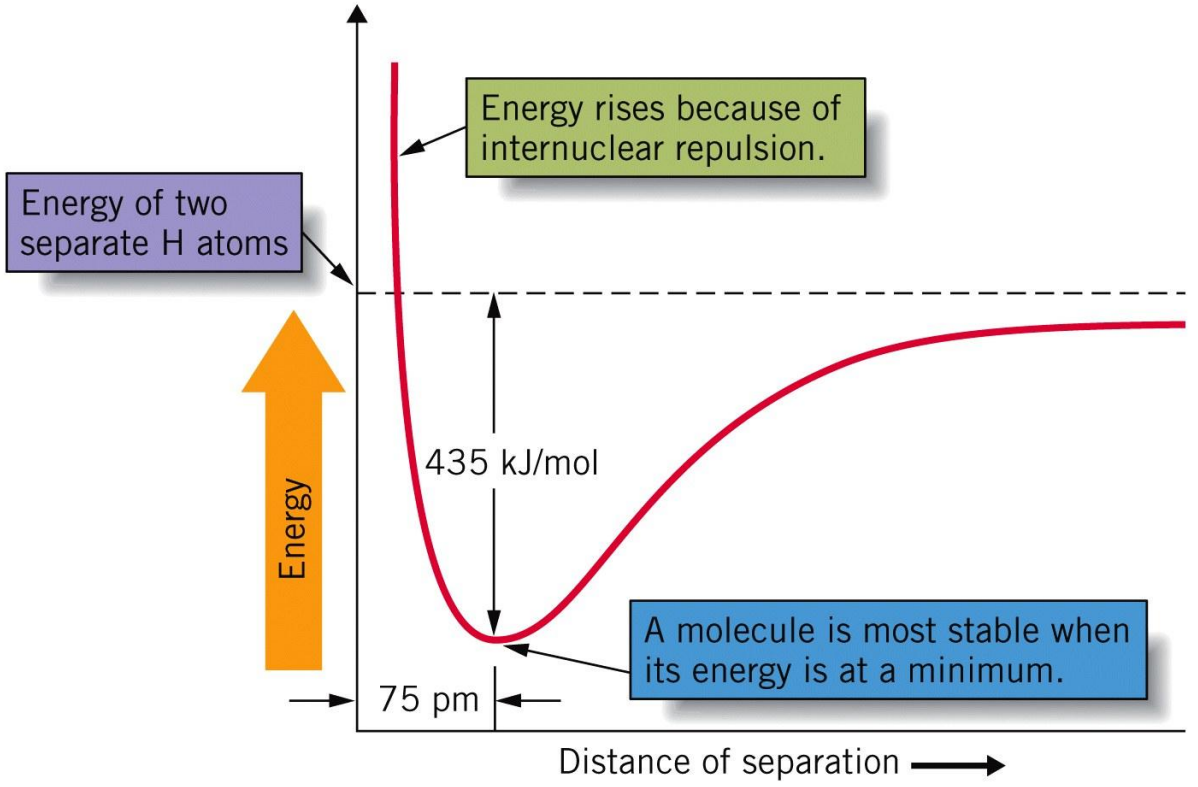
- Attraction of valence $e^-$ of 1 atom by nucleus of other atom
- Shifting of $e^-$ density
- As distance between nuclei ↓, ↑ probability of finding either $e^-$ near either nucleus
- Pulls nuclei closer together




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# Covalent Bond

- As nuclei get close
  - Begin to repel each other
  - Both have high positive charge



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- Final internuclear distance between 2 atoms in bond
  - Balance of attractive and repulsive forces
  - Net attraction since bond forms

# Electron Pair Bond

- $e^-$'s come together in molecular bond
- In sense **1*s*** orbital of each H is filled
- $e^-$'s share same region of space ∴ their spins must be paired (1 ↑ and 1 ↓)
- Refer to as ***$e^-$ pair bond***

http://www.visionlearning.com/library/module_viewer.php?c3=&mid=55&l=

# Bond Dissociation Energy

- Energy needed to break one mole of bond in covalent molecule in gas phase
- Symbolized **D**
  - **D** varies with bond type
    - Single, double, triple
  - **D** varies with atoms involved
  - **D** is always positive
- Formation of bonds stabilizes structure energetically

# Chemical Potential Energy

$$\Delta H^\circ_{rxn} = \Sigma\{D_{(Bonds\ broken)}\} - \Sigma\{D_{(Bonds\ formed)}\}$$

| Bond | Bond Energy (kJ mol−1) | Bond | Bond Energy (kJ mol−1) |
|---|---|---|---|
| C–C | 348 | C–Br | 276 |
| C═C | 612 | C–I | 238 |
| C≡C | 960 | H–H | 436 |
| C–H | 412 | H–F | 565 |
| C–N | 305 | H–Cl | 431 |
| C═N | 613 | H–Br | 366 |
| C≡N | 890 | H–I | 299 |
| C–O | 360 | H–N | 388 |
| C═O | 743 | H–O | 463 |
| C–F | 484 | H–S | 338 |
| C–Cl | 338 | H–Si | 376 |

# Covalent Bond

- **2 quantities characterize this bond**

**Bond Length** (bond distance)

- Distance between 2 nuclei = $r_A + r_B$

**Bond Energy**

- Also ***bond strength***
- Amount of energy released when bond formed (↓ PE) or
- Amount of energy must put in to “break” bond

# Your Turn!

Which species is most likely covalently bonded?

A. CsCl

B. NaF

C. $CaF_2$

D. CO

E. $MgBr_2$

# Lewis Structures

- Molecular formula drawn with Lewis Symbols
- Method for diagramming electronic structure of covalent bonds
- Uses dots to represent $e^-$s
- Covalent bond
  - Shared pair of $e^-$s
  - Each atom shares $e^-$s so has complete octet $ns^2np^6$
    - Noble Gas $e^-$ configuration
    - Except H which has complete shell with 2 $e^-$s

# Octet Rule

When atoms form covalent bonds, tend to share sufficient $e^-$'s so as to achieve outer shell having 8 $e^-$'s

- Indicates how all atoms in molecule are attached to one another
- Accounts for ALL valence $e^-$'s in ALL atoms in molecule

Let's look at some examples

**Noble Gases:** 8 valence $e^-$'s

- Full octet $\boldsymbol{ns^2\ np^6}$
- Stable monatomic gases
- Don't form compounds

# Lewis Structures

## Diatomic Gases:

- H and Halogens

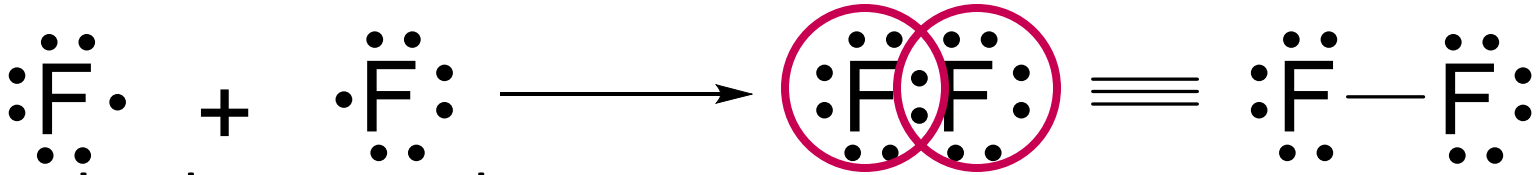
  $\mathbf{H_2}$

- H· + ·H → H:H or H—H
- Each H has 2 e⁻'s through sharing
- Can write shared pair of e⁻'s as line (—)
- : = —
- covalent bond

# Lewis Structures

## Diatomic Gases:

$F_2$

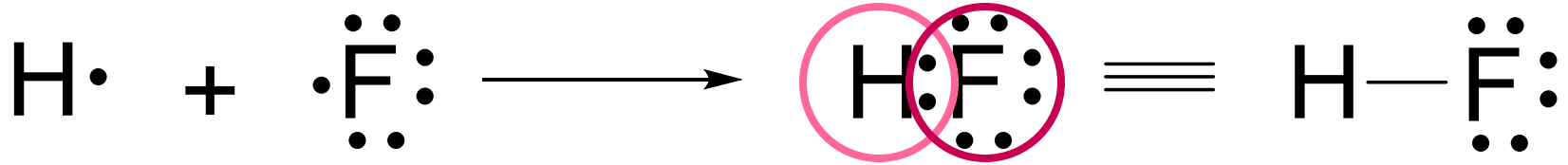


- Each F has complete octet
- Only need to form one bond to complete octet
- Pairs of $e^-$'s not included in covalent bond are called Lone Pairs
- Same for rest of Halogens: $Cl_2$, $Br_2$, $I_2$

# Lewis Structures

## Diatomic Gases:

**HF**



- Same for HCl, HBr, HI.
- Molecules are diatomics as need only $1e^-$ to complete octet
- Separate molecules
  - Gas in most cases because very weak intermolecular forces

# Your Turn!

How many electrons are required to complete the octet around nitrogen, when it forms $N_2$ ?

A. 2

B. 3

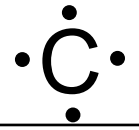
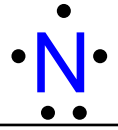
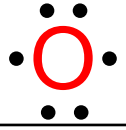
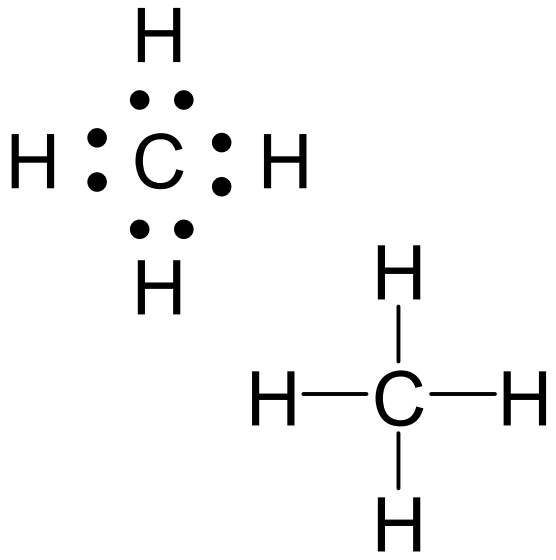
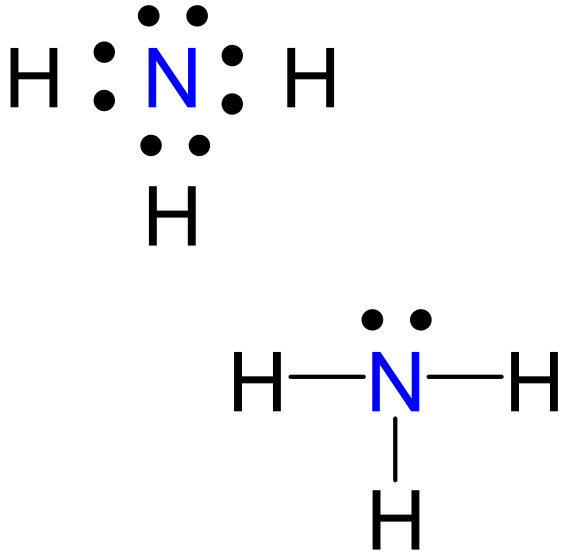
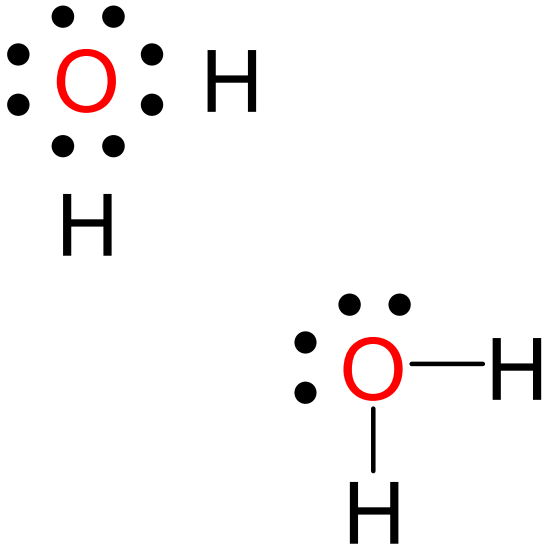
C. 1

D. 4

E. 6

# Lewis Structures

- Many nonmetals form more than 1 covalent bond

|  |  |  |
|---|---|---|
| Needs 4 e⁻'s<br>Forms 4 bonds | Needs 3 e⁻'s<br>Forms 3 bonds | Needs 2 e⁻'s<br>Forms 2 bonds |
|  |  |  |
| **methane** | **ammonia** | **water** |

# Multiple Bonds

## Single Bond

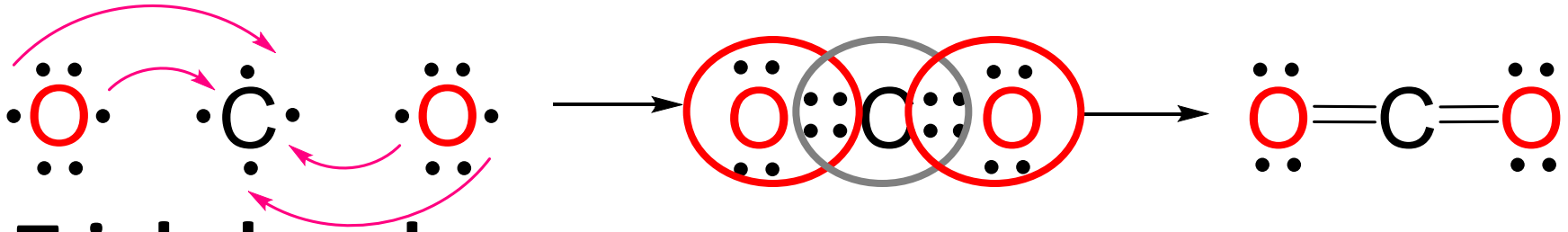
- Bond produced by sharing ***one pair*** of $e^-$'s between 2 atoms

- Many molecules share *more than one pair* of $e^-$'s between 2 atoms
  - ***Multiple*** bonds

# Double Bonds

- 2 pairs of e⁻'s shared between 2 atoms

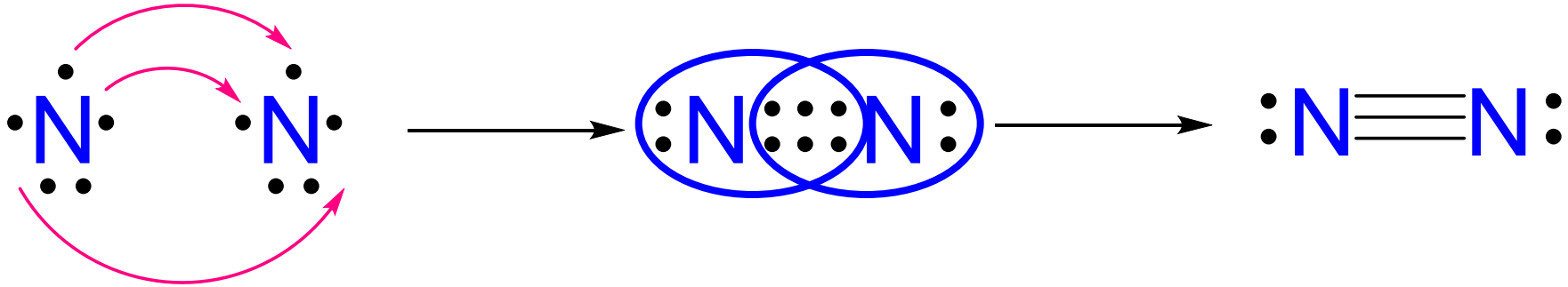
**Ex.** $CO_2$




## Triple bond

- 3 pairs of e⁻'s shared between 2 atoms

**Ex.** $N_2$



# Your Turn!

Which species is most likely to have multiple bonds ?

A. CO

B. $H_2O$

C. $PH_3$

D. $BF_3$

E. $CH_4$

# Carbon Compounds

- Carbon-containing compounds
  - Exist in large variety
  - Mostly due to multiple ways in which C can form bonds
- **Functional groups**
  - Groups of atoms with similar bonding
  - Commonly seen in C compounds
- Molecules may contain more than 1 functional group

# Important Compounds of Carbon

**Alkanes**

- Hydrocarbons
- Only single bonds

**Isomers**

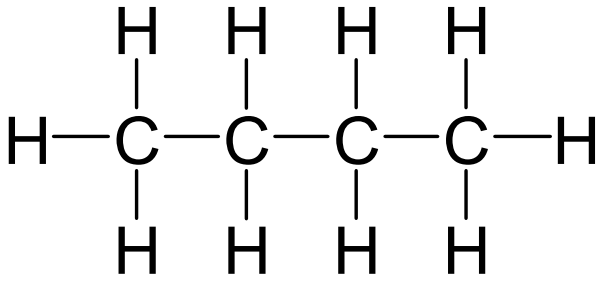
- Same molecular formula
- Different physical properties
- Different connectivity (structure)

Ex.

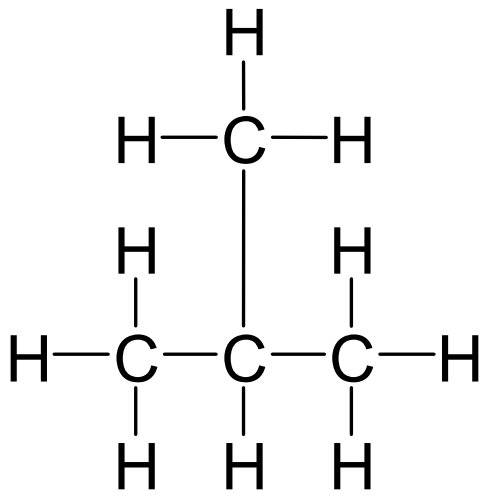
$CH_4$ methane

$CH_3CH_3$ ethane

$CH_3CH_2CH_3$ propane



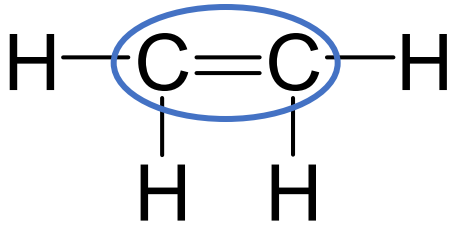
butane



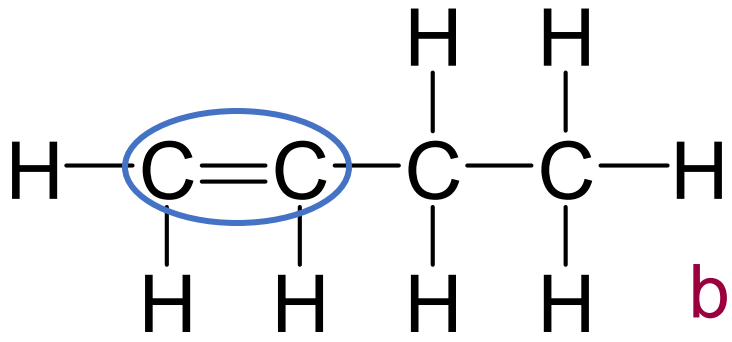
iso-butane

# Hydrocarbons

- **Alkenes**
  - Contain at least one double bond

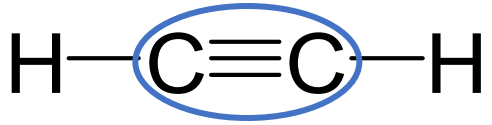


Ethylene
(ethene)

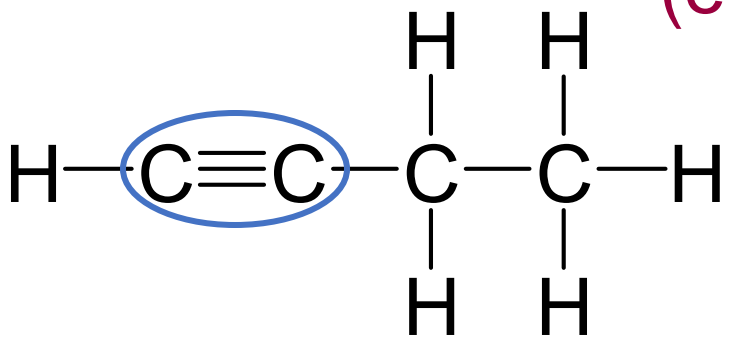


butene

- **Alkynes**
  - Contain at least one triple bond



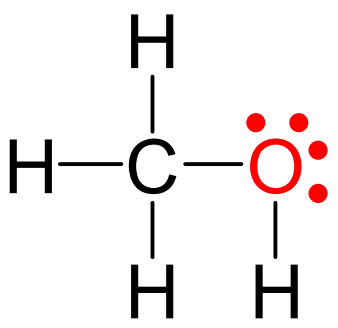
acetylene
(ethyne)



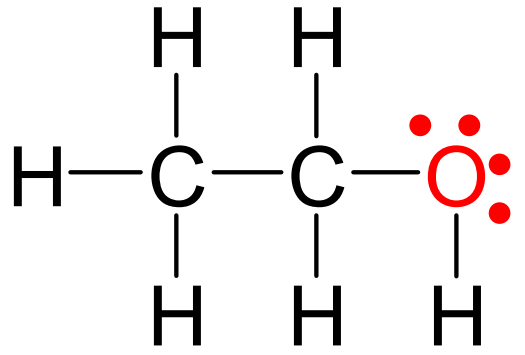
butyne

# Oxygen Containing Organics

- **Alcohols**
  - Replace H with OH

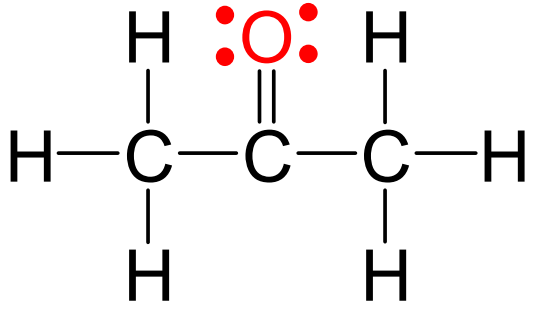


methanol



ethanol

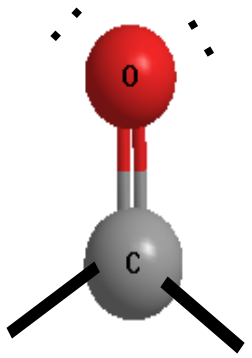
- **Ketones**
  - Replace $CH_2$ with C=O
  - **Carbonyl** group



acetone

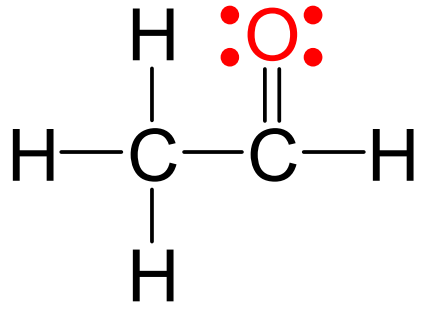
# Carbonyl Group

- Carbon in hydrocarbon with double bond to oxygen
  - Aldehydes
  - Ketones
  - Carboxylic acids
  - Amides

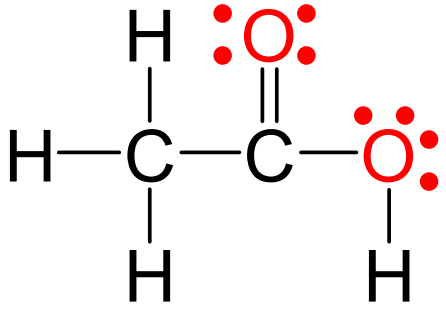


# Oxygen Containing Organics

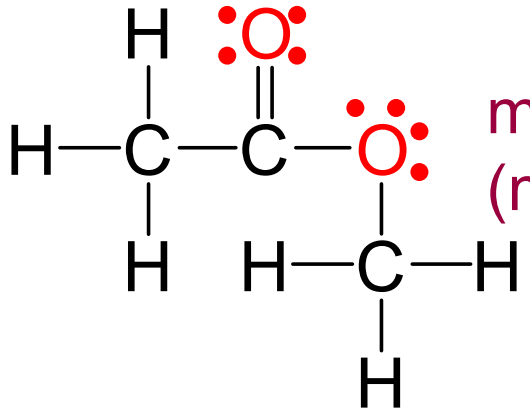
- **Aldehydes**
  - At least 1 atom attached to C=O is H
- **Organic Acids**
  - Contains **carboxyl** group
  - COOH
- **Esters**
  - Alkyl group replaces H on carboxylic acid



acetaldehyde



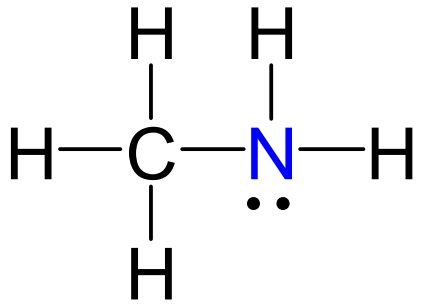
acetic acid



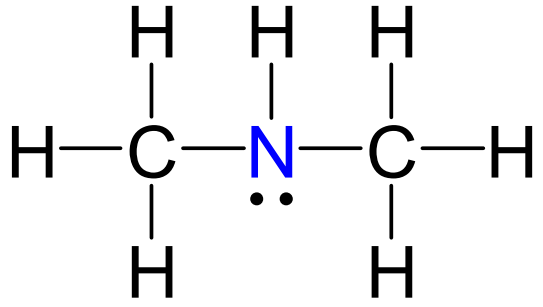
methyl acetate
(methylethanoate)

# Nitrogen Containing Organics

- **Amines**
  - derivatives of $NH_3$ with H's replaced by alkyl groups

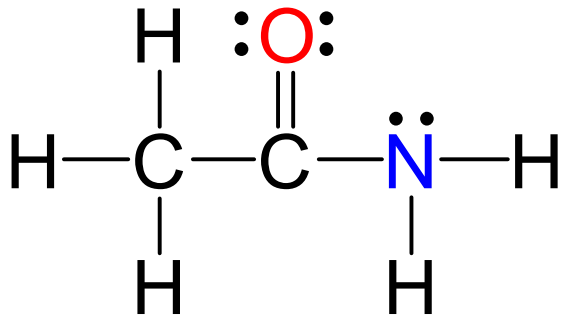


methylamine



dimethylamine

- Amides
  - Contain -$NH_2$ on carbon chain with carbonyl



acetamide (ethanamide)

# Your Turn!

How many isomers are there of butanol?

A. none

B. 2

C. 3

D. 6

E. 4

# Drawing Lewis Structures

- Very useful
- Way of diagramming structure
- Used to describe structure of molecules
- Can be used to make reasonable accurate predictions of shapes of molecules

# Drawing Lewis Structures

- Not all molecules obey the Octet Rule.
  - Holds rigorously for 2nd row elements like C, N, O, and F
  - B and Be sometimes have less than octet $BeCl_2$, $BCl_3$
  - 2nd row can **never** have **more** than 8 e⁻'s
  - 3rd row and below, atoms often **exceed** octet
- **Why?**
  - ***n*** = 3 shell can have up to 18 e⁻'s as now have ***d*** orbitals in valence shell

# Method for Drawing Lewis Structures

1. Decide how atoms are bonded
   - **Skeletal structure** = arrangement of atoms.
   - **Central atom**
     - Usually given first
     - Usually least electronegative
2. Count all **valence** $e^-$s. (All atoms)
3. Place 2 $e^-$s between each pair of atoms
   - Draw in single bonds

# Method for Drawing Lewis Structures

4. Complete octets of **terminal atoms** (atoms attached to central atom) by adding $e^-$s in pairs
5. Place any remaining $e^-$'s on central atom in pairs
6. If central atom does not have octet
   - Form double bonds
   - If necessary, form triple bonds

# Ex. $N_2F_2$

2 N = 2 × 5e⁻ = 10 e⁻

2F = 2 × 7e⁻ = 14 e⁻

Total = 24 e⁻

single bonds – 6 e⁻

18 e⁻

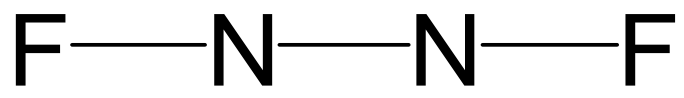
F lone pairs – 12 e⁻

6 e⁻

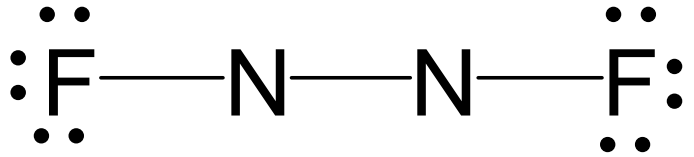
N electrons – 6 e⁻

0 e⁻

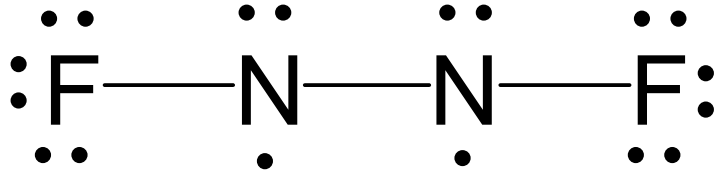
Skeletal Structure




Complete terminal atom octets

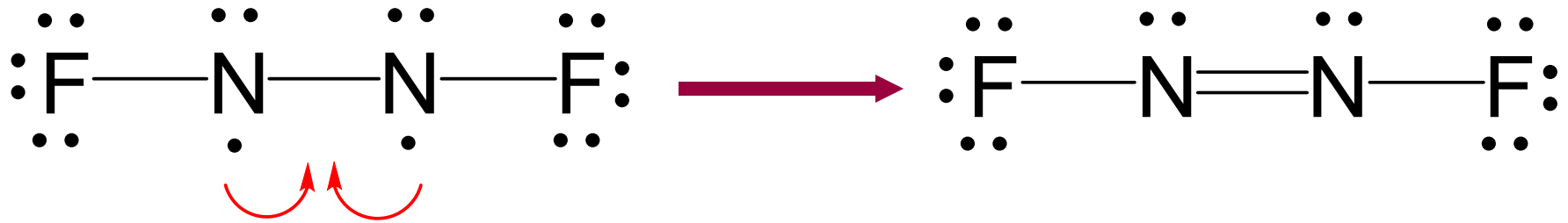



Put remaining e⁻s on central atom



# Ex. $N_2F_2$

- Not enough electrons to complete N octets
- Must form double bond between N's to satisfy both octets.



# Ex. $SiF_4$

1 Si = 1 × 4e⁻ = 4 e⁻

4F = 4 × 7e⁻ = 28 e⁻

Total = 32 e⁻

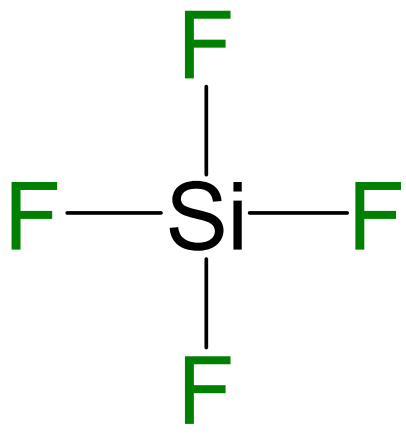
single bonds – 8 e⁻

24 e⁻

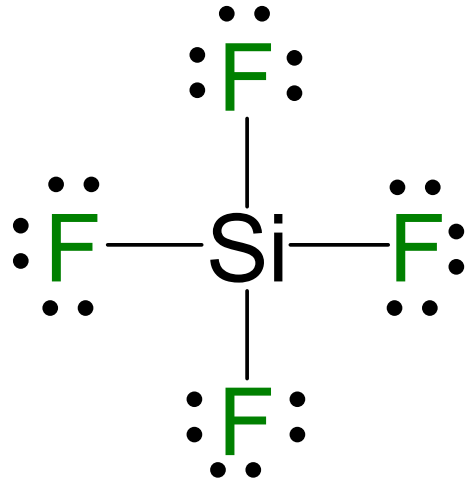
F lone pairs – 24 e⁻

0 e⁻

Skeletal Structure




Complete terminal atom octets



# Ex. $H_2CO_3$

- $CO_3^{2-}$ oxoanion, so C central, and O's around, $H^+$ attached to two O's

1 C = 1 × 4e⁻ = 4 e⁻

3 O = 3 × 6e⁻ = 18 e⁻

2 H = 2 × 1e⁻ = 2 e⁻

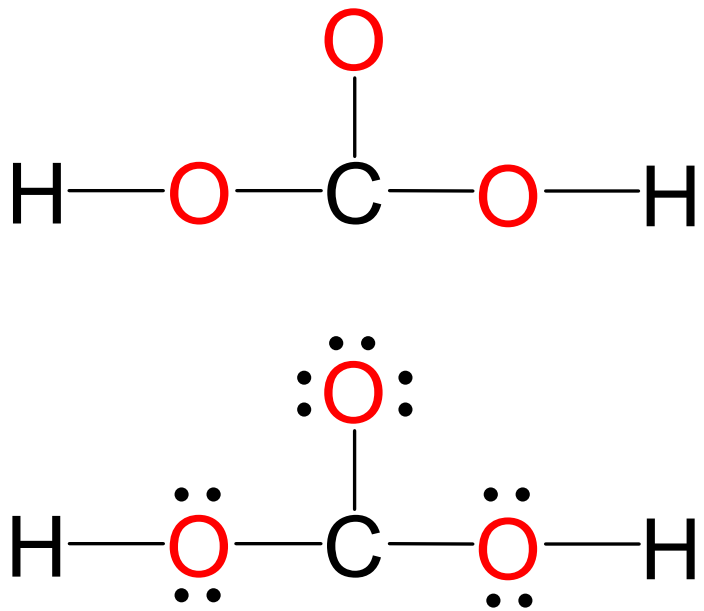
Total = 24 e⁻

single bonds – 10 e⁻

14 e⁻

O lone pairs – 14 e⁻

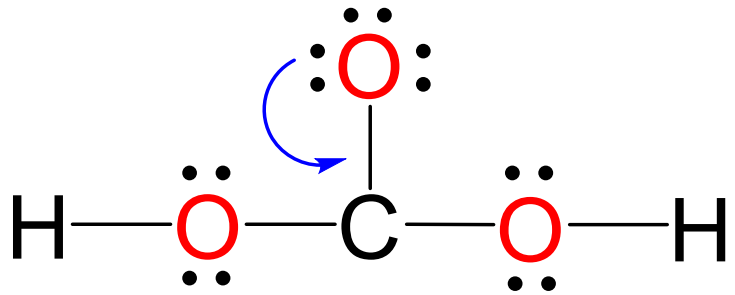
0 e⁻

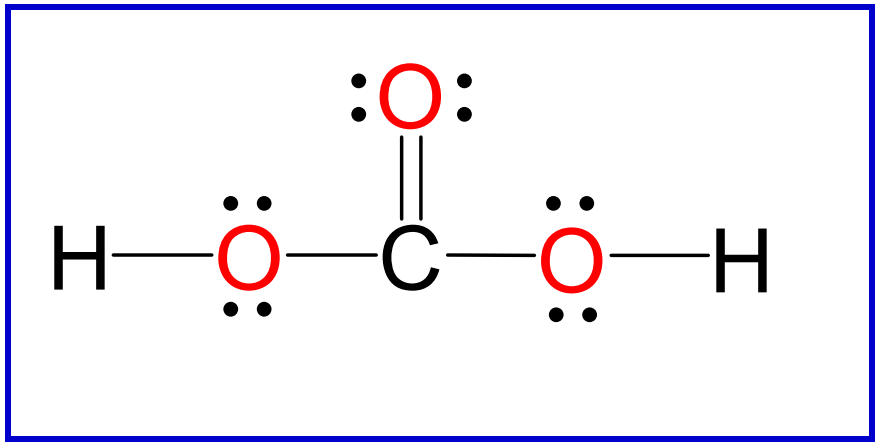



- But C only has 6 e⁻

# Ex. $H_2CO_3$ (cont.)

- Too few $e^-$
- Must convert 1 of lone pairs on O to 2nd bond to C
- Form double bond between C and O



# Ex. $PCl_5$

1 P = 1 × 5 $e^-$ = 5 $e^-$

5 Cl = 5 × 7$e^-$ = 35 $e^-$

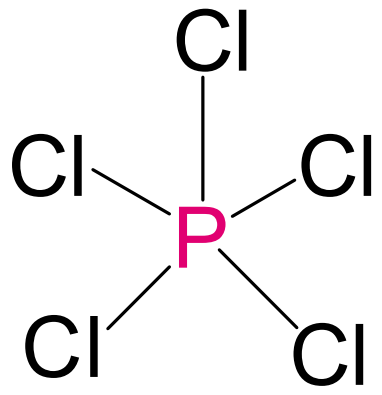
Total = 40 $e^-$

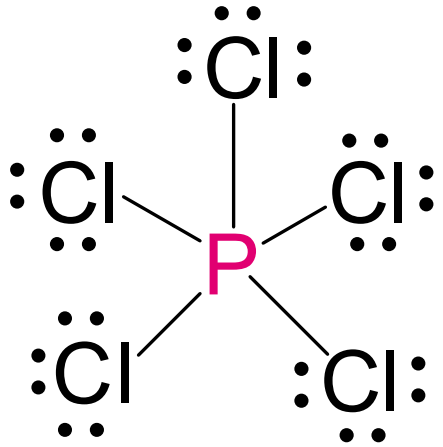
single bonds – 10 $e^-$

30 $e^-$

Cl lone pairs – 30 $e^-$

0 $e^-$




- P has 10 $e^-$
- OK as 3rd row element
- Can *expand* its shell

# Ex. $BBr_3$

1 B = 1 × 3 $e^-$ = 3 $e^-$

3 Br = 3 × 7$e^-$ = 21 $e^-$

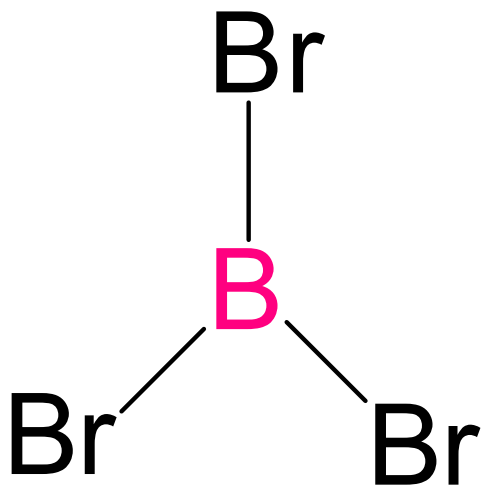
Total = 24 $e^-$

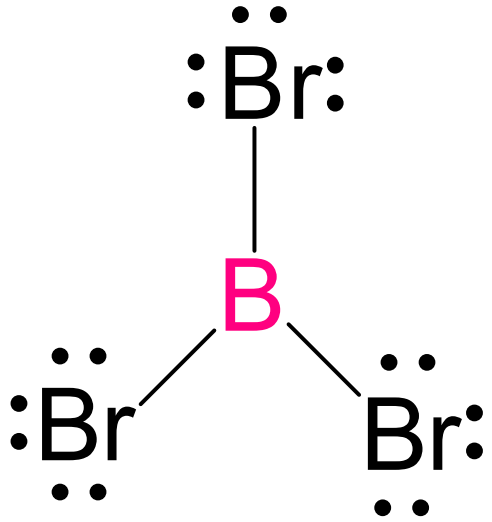
single bonds – 6 $e^-$

18 $e^-$

Br lone pairs – 18 $e^-$

0 $e^-$




- B has only 6 $e^-$
- Does *not* form double bond
- Has *incomplete* octet

# Ex. $CO_2$

1 C = 1 × 4e⁻ = 4 e⁻

2 O = 2 × 6e⁻ = 12 e⁻

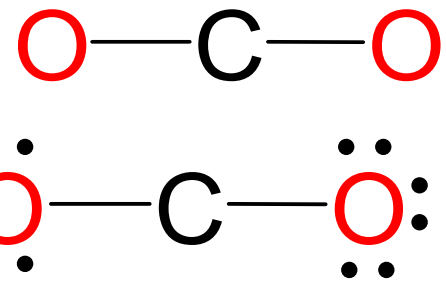
Total = 16 e⁻

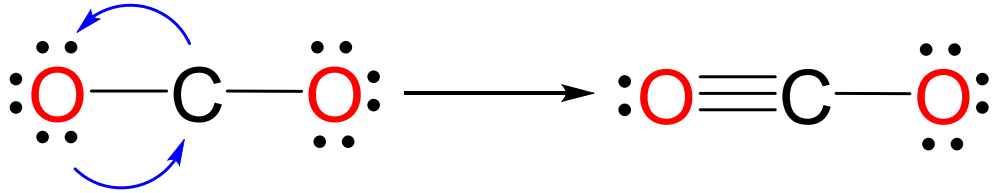
single bonds – 4 e⁻

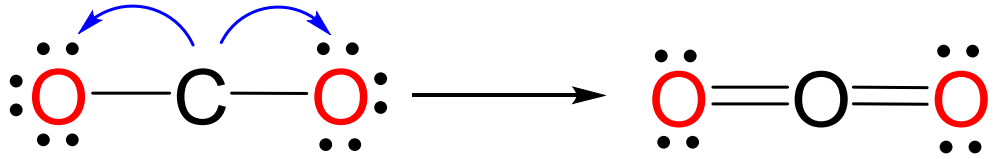
12 e⁻

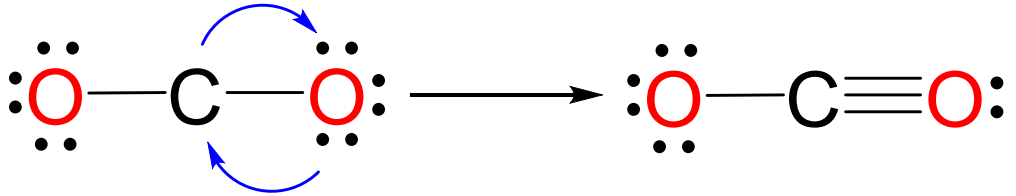
O lone pairs – 12 e⁻

0 e⁻









- C only has 4 e⁻
- Must form 2 double bonds to O to complete C's octet
- 3 ways you can do this

- Which of these is correct?
- Need another criteria
- Come back to this

# Lewis Structures - Theory

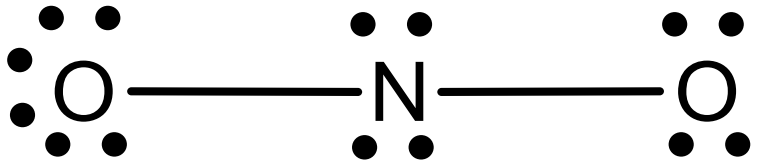
- Lewis Structures meant to describe how **atoms share** $e^-$s in chemical bonds
  - Theory

**What experimental observations back up this theory?**

- Look at properties related to number of $e^-$ pairs shared between 2 atoms

# Your Turn!

What is wrong with the following structure?




A. Too few total electrons

B. Too many total electrons

C. Lack of octet around nitrogen

D. Too many electrons around the N atom

E. Structure is correct as written

# Review of Covalent Bond Properties

**Bond Length**

- Distance between nuclei of bonded atoms

**Bond Energy**

- Energy required to separate bonded atoms into neutral particles

**Bond Order**

- Number of pairs of $e^-$s shared between 2 atoms
- Measure of amount of $e^-$ density in bond

# Bond Order, Bond Length and Bond Energy

**Greater Bond Order** = Greater $e^-$ density

- Nuclei held together more tightly
- **Larger bond energy**, D

- Larger D means
  - Nuclei drawn closer together
  - **Shorter bond length**
- **As bond order ↑, bond length ↓, and bond energy ↑**
  - Assumes comparing bonds between same 2 elements

# Ex. $H_2SO_4$

1 S = 1 × 6e⁻ = 6 e⁻

4 O = 4 × 6e⁻ = 24 e⁻

2 H = 2 × 1e⁻ = 2 e⁻

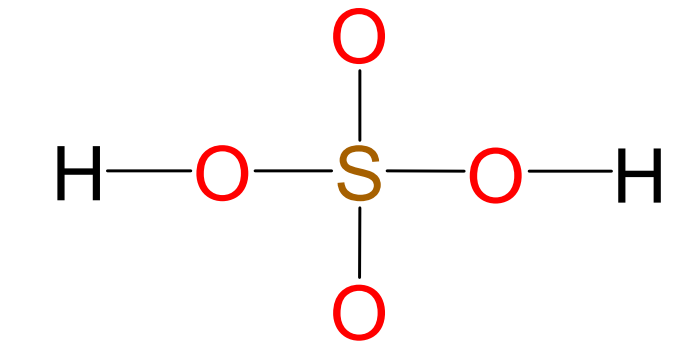
Total = 32 e⁻

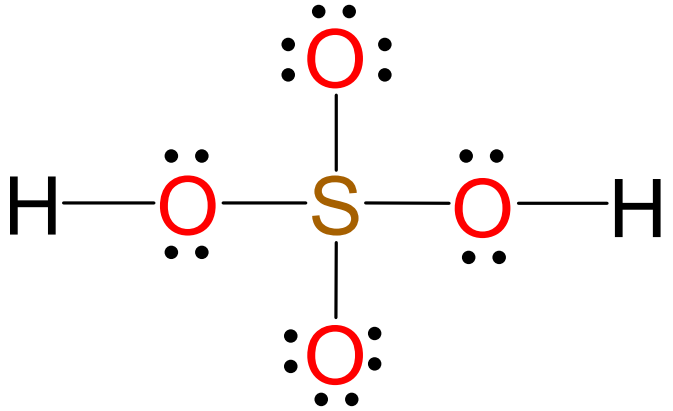
single bonds – 12 e⁻

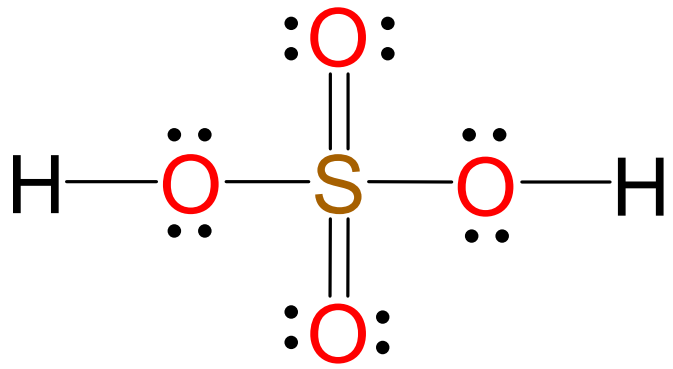
20 e⁻

O lone pairs – 20 e⁻

0 e⁻







- *n*=3, has empty *d* orbitals
- Could *expand* it's octet
- Could write structure with double bonds.

# How Do We Know Which is Accurate

## Experimental evidence

- In this case bond lengths from X-ray data
- S—O bonds (no H attached) are shorter 142 pm
- S—O—H, S—O longer 157 pm
- Indicates that 2 bonds are shorter than other 2
- Structure with S=O for 2 O's without H's is more accurate
- Preferred Lewis Structure
  - Even though it seems to violate octet rule unnecessarily

# Need Criteria to Predict When This Occurs

- Covalent bond = shared $e^-$ pair
- If $e^-$s shared equally
  - Each atom "owns" ½ of **e**$^-$ pair or 1 **e**$^-$
- When "break" bond
  - ½ of these **e**$^-$s going to each atom in bond
- So for S
  - have 4 bonds
  - Have 4 **e**$^-$s
  - 2 **e**$^-$s less than 6 **e**$^-$s, S has as atom
- In bookkeeping sense
  - S has 2+ charge if it obeys octet rule

# Formal Charge

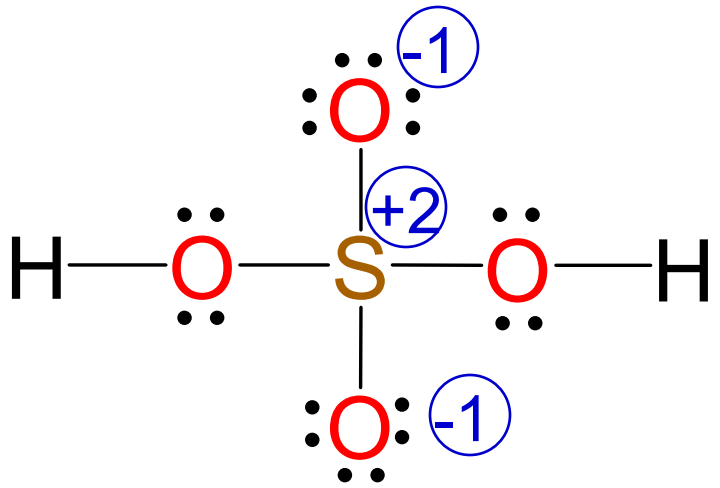
- Apparent charge on atom
- Bookkeeping method
- Does not represent real charges

**FC = # valence $e^-$ – # lone pair $e^-$ – ½ (# bonding $e^-$)**

**FC = # valence $e^-$ – [# bonds to an atom + # unshared $e^-$ ]**

- Indicate Formal charges by placing them in circles around atoms

FC = #valence $e^-$ – [#bonds to atom + # unshared $e^-$ ]



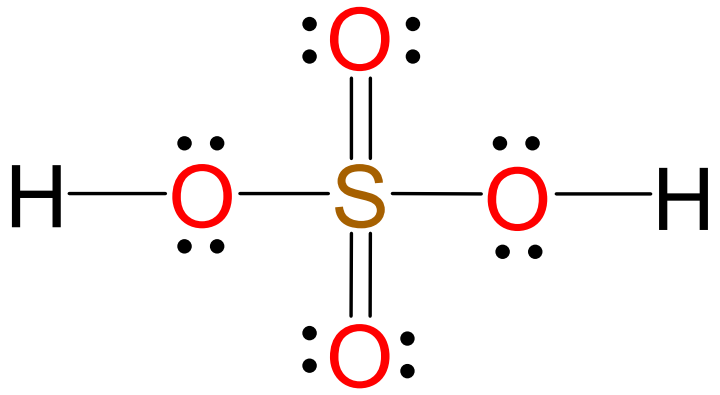

**Structure 1**

$FC_S = 6 - (4 + 0) = 2$

$FC_H = 1 - (1 + 0) = 0$

$FC_{O(s)} = 6 - (1 + 6) = -1$

$FC_{O(d)} = 6 - (2 + 4) = 0$




**Structure 2**

$FC_S = 6 - (6 + 0) = 0$

$FC_H = 1 - (1 + 0) = 0$

$FC_{O(s)} = 6 - (2 + 4) = 0$

$FC_{O(d)} = 6 - (2 + 4) = 0$

# $H_2SO_4$

- No formal charges on any atom in Structure 2

**Conclusion:**

- When several Lewis Structures are possible
- Those with **smallest** formal charges
  - Most stable
  - Preferred

## Most Stable Lewis Structure

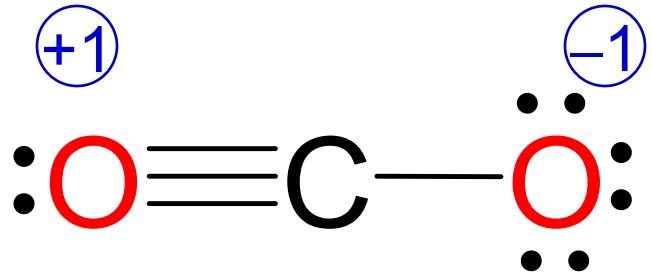
1. Least number of FC's best
2. All FC $\leq$ |1|
3. Any negative FC on most electronegative element

# Can Use Formal Charge to Explain Boron Chemistry

- $BCl_3$
- Why doesn’t a double bond form here?
- $FC_B = 3 – 0 – 3 = 0$
- $FC_{Cl} = 7 – 6 – 1 = 0$
- All FC's = 0 so stable, doesn't need to form double bond

# $CO_2$ which structure is best

- Use FC to determine which structure is best

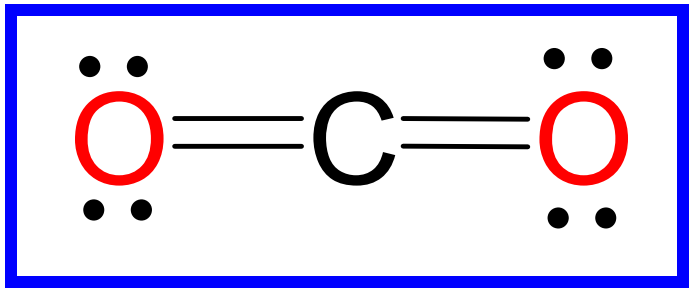



$FC_C = 4 - (4 + 0) = 0$

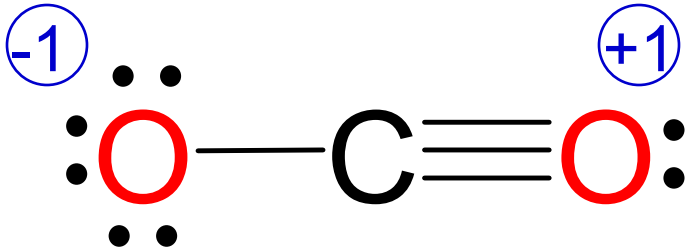
$FC_{O(s)} = 6 - (1 + 6) = -1$

$FC_{O(d)} = 6 - (2 + 4) = 0$

$FC_{O(t)} = 6 - (3 + 2) = +1$

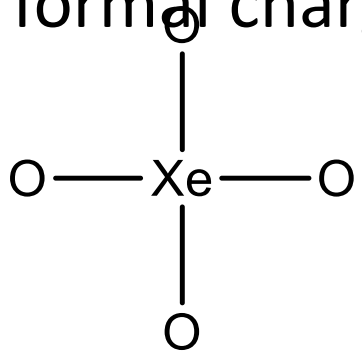


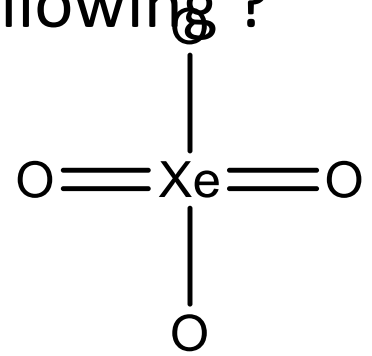
- Central structure Best
- All FC's = 0



# Your Turn!

What is the formal charge on Xe for the following ?




A. +2, +4

B. +2, +3

C. +4, 0

D. +4, +2

# Resonance: When Single Lewis Structure Fails

$\chi_N = 3$ $\chi_O = 3.5$

1 N = 1 × 5e⁻ = 5 e⁻

3 O = 3 × 6e⁻ = 18 e⁻

–1 charge = 1 e⁻

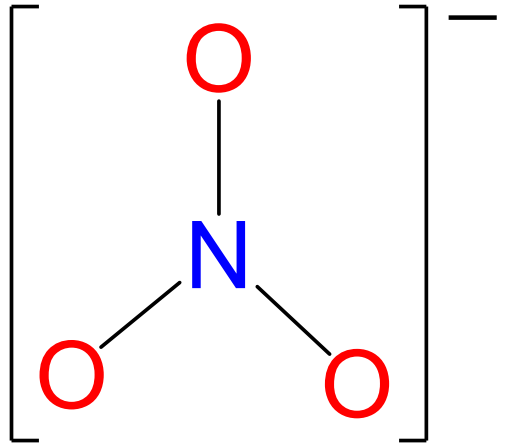
Total = 24 e⁻

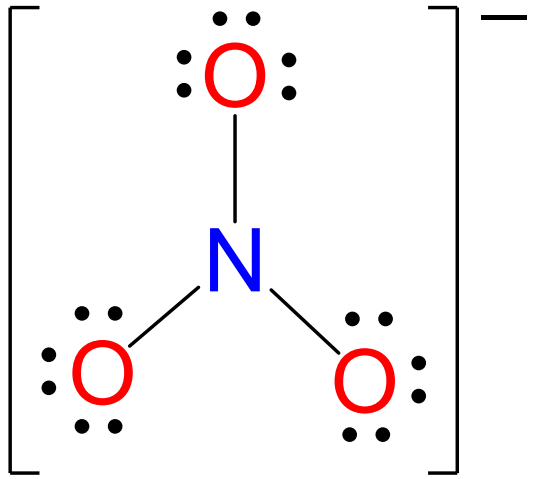
single bonds – 6 e⁻

18 e⁻

O lone pairs –18 e⁻

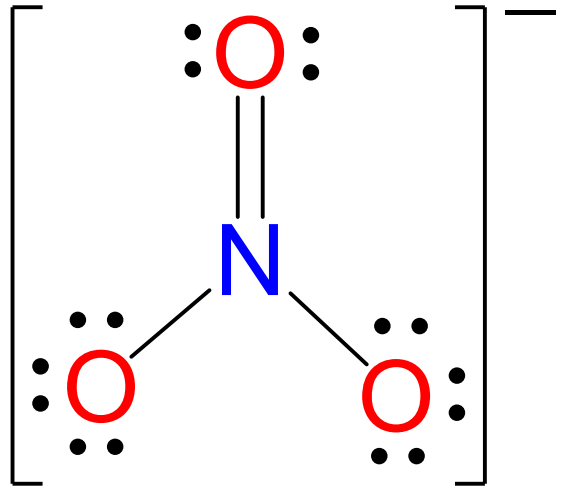
0 e⁻



# Ex. $NO_3^-$

- Lewis structure predicts 1 bond shorter than other 2



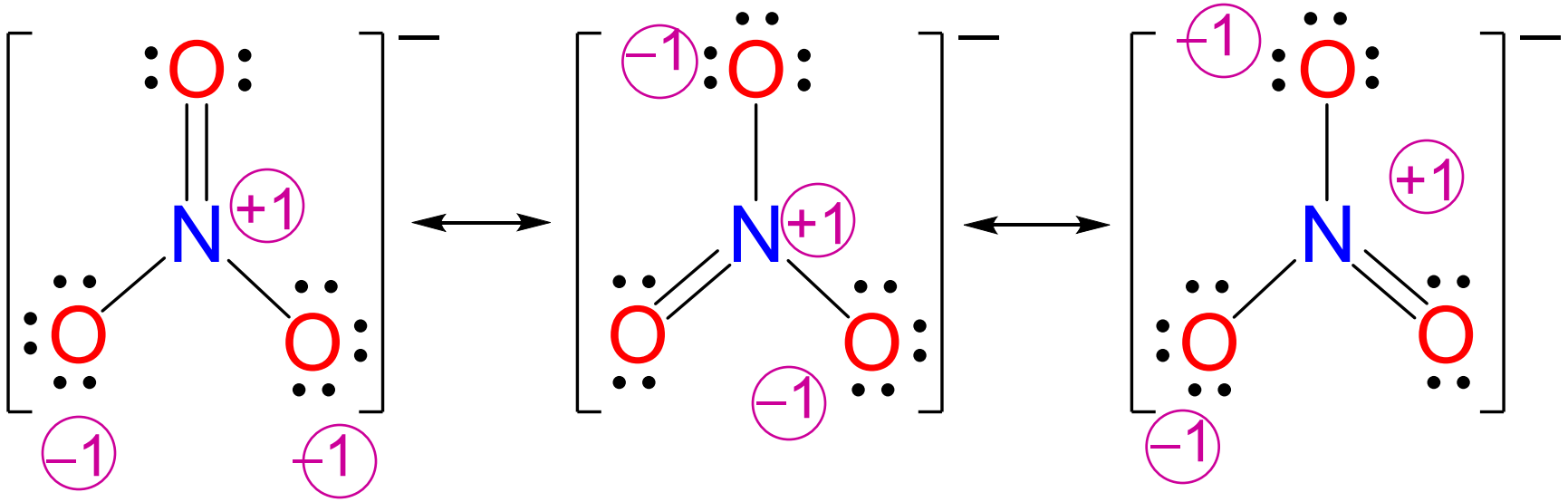

Experimental observation:

- All 3 N—O bond lengths are same
- All shorter than N—O single bonds

- Have to modify Lewis Structure
  - $e^-$ can't distinguish O atoms
- Can write 2 or more possible structures simply by moving where $e^-$ are
  - Changing placement of $e^-$

# What are Resonance Structure ?

- Multiple Lewis Structures for single molecule
  - No single Lewis structure is correct
  - Structure not accurately represented by any 1 Lewis Structure
  - Actual structure = "average" of all possible structures
  - Double headed arrow between resonance structures used to denote resonance

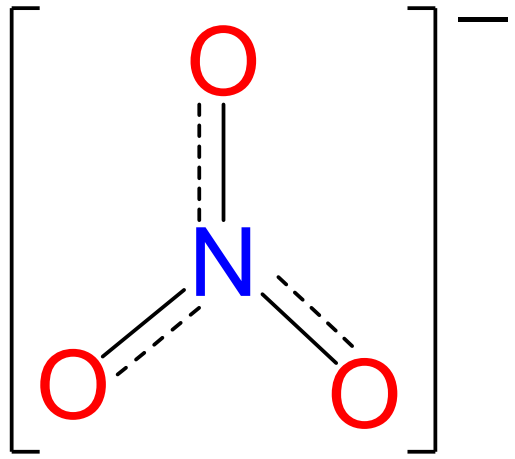


# Resonance Structure

- Lewis structures assume **e**$^-$ is localized between 2 atoms
- In cases where need resonance structures, **e**$^-$ are **delocalized**
  - Smeared out over all atoms
  - Can move around entire molecule to give equivalent bond distances

**Resonance Hybrid**

- Way to depict resonance delocalization



# Ex. $CO_3^{2-}$

$\chi_C = 2.5$ $\chi_O = 3.5$

1 N = 1 × 5e⁻ = 5 e⁻

3 O = 3 × 6e⁻ = 18 e⁻

–1 charge = 1 e⁻

Total = 24 e⁻

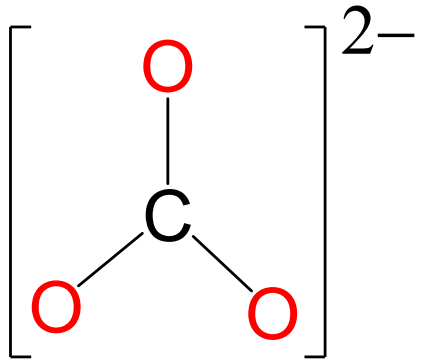
single bonds – 6 e⁻

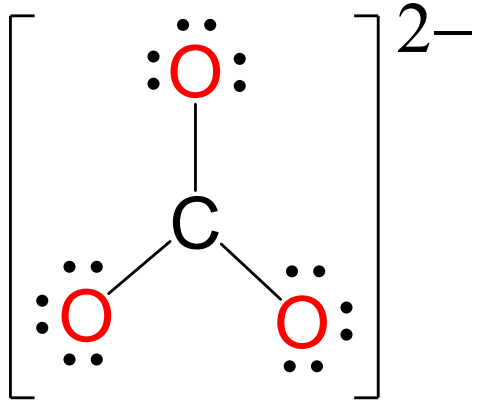
18 e⁻

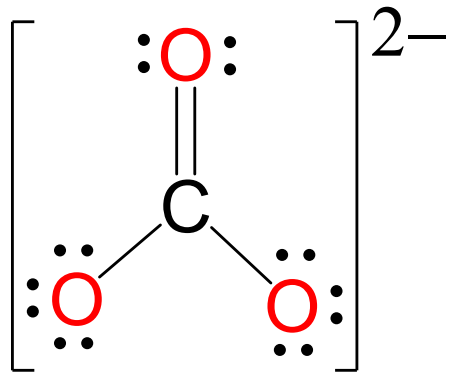
O lone pairs –18 e⁻

0 e⁻

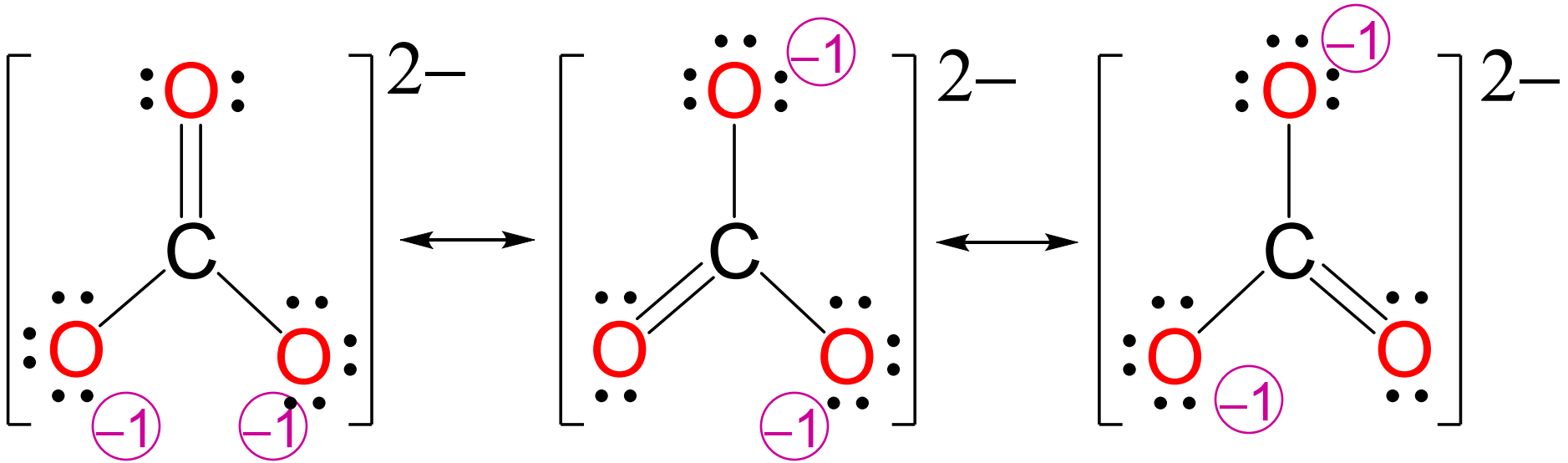
C only has 6 e⁻ so form double bond



# Resonance Structure




- All have same net formal charges on C and O's
  - FC = –1 on singly bonded O's
  - FC = O on doubly bond O and C

# Resonance Structure not always equivalent

- 2 or more Lewis Structures for **same** compound may or may not represent $e^-$ distributions of equal energy

**How Do We Determine Which are Good Contributors?**

1. All Octets are satisfied
2. All atoms have as many bonds as possible

**3a.** FC $\leq |1|$

**3b.** Any negative charges are on electronegative atoms.

4. As little charge separation as possible.

# Draw Good Resonance Structure

1. All must be valid Lewis Structures
2. Only $e^-$ are shifted
   - Usually double or triple bond and lone pair
   - Nuclei can't be moved
   - Bond angles must remain the same
3. Number of unpaired electrons, if any, must remain the same
4. Major contributors are the ones with lowest E (see above)
5. Resonance stabilization is most important when serves to delocalize charge onto 2 or more atoms

# Ex. $NCO^-$

$\chi_N = 3$ $\chi_C = 2.5$ $\chi_O = 3.5$

1 C = 1 × 4e⁻ = 4 e⁻

1 N = 1 × 5e⁻ = 5 e⁻

1 O = 1 × 6e⁻ = 6 e⁻

–1 charge = 1 e⁻

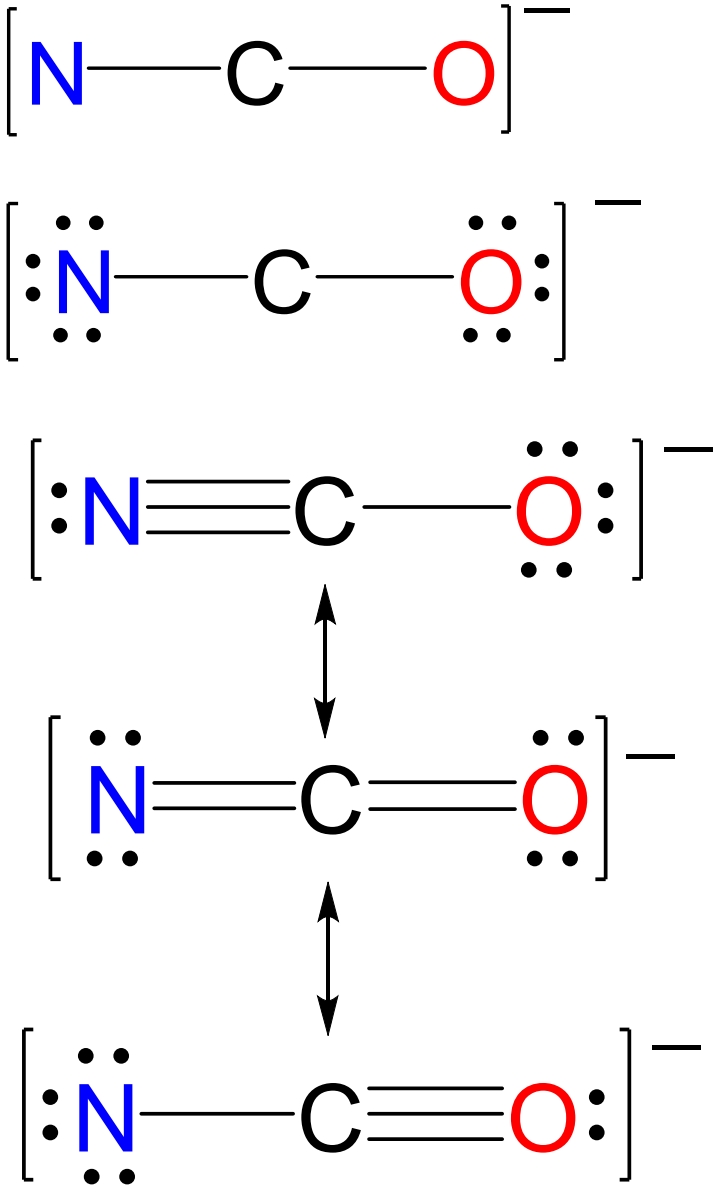
Total = 16 e⁻

single bonds – 4 e⁻

12 e⁻

O lone pairs –12 e⁻

0 e⁻



# Ex. $NCO^-$

$FC_N = 5 - 2 - 3 = 0$

$FC_C = 4 - 0 - 4 = 0$

$FC_O = 6 - 6 - 1 = -1$

$FC_N = 5 - 4 - 2 = -1$

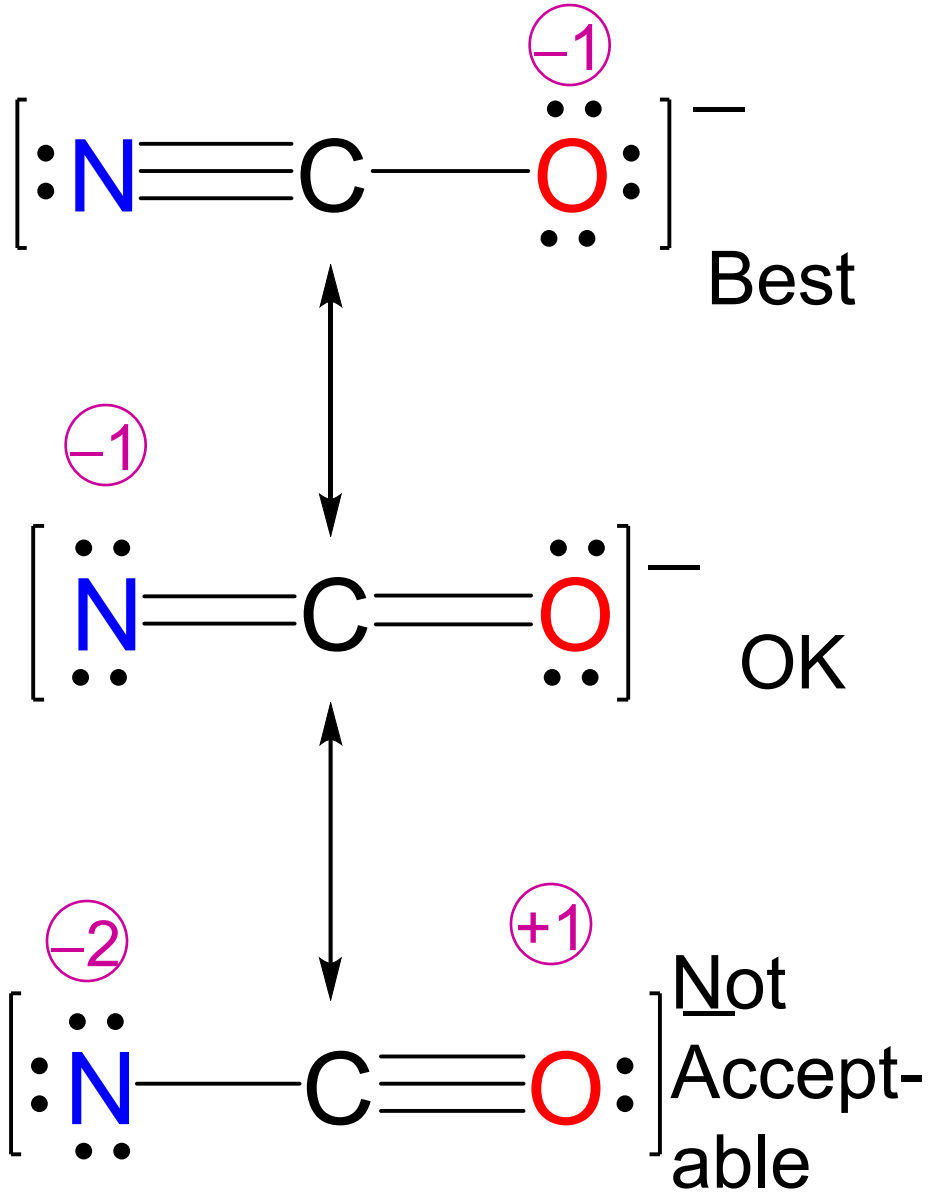
$FC_C = 4 - 0 - 4 = 0$

$FC_O = 6 - 4 - 2 = 0$

$FC_N = 5 - 6 - 1 = -2$

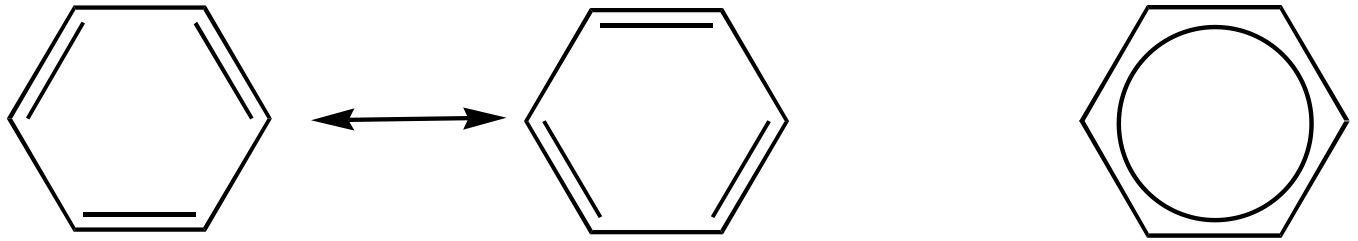
$FC_C = 4 - 0 - 4 = 0$

$FC_O = 6 - 2 - 3 = +1$



# Resonance Stabilization

- Actual structure is more stable than either single resonance structure





- For Benzene
- The extra stability is ~146 kJ/mol
- **Resonance Energy**
  - Extra stabilization energy from resonance

# Your Turn!

Which of the structures below exhibit resonance?

A. $NO_2$

B. $H_2O$

C. $N_3^-$

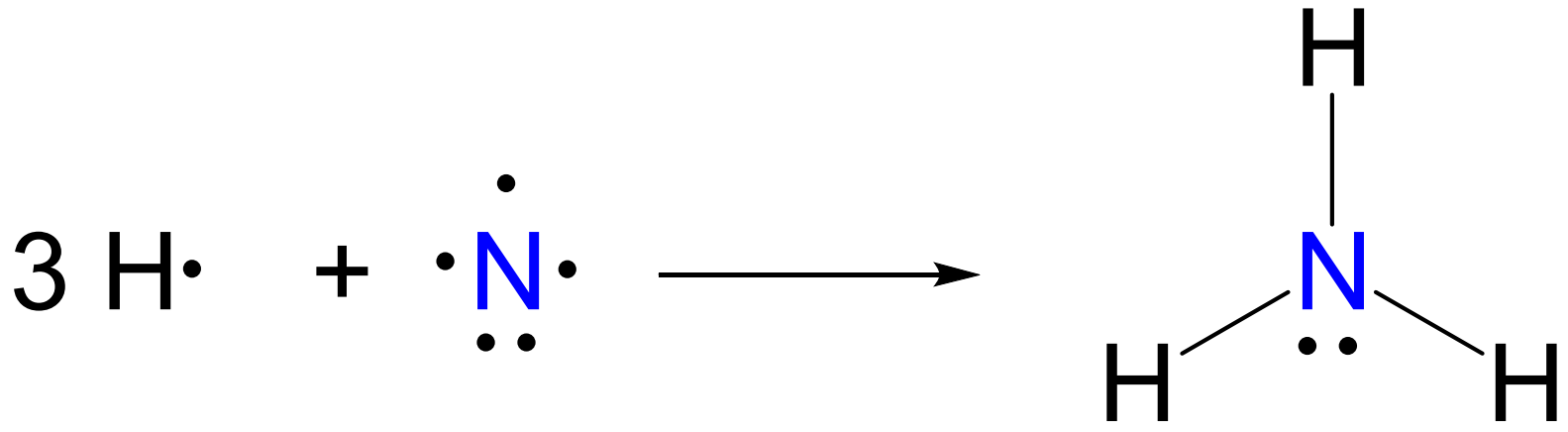
D. $N_2O$ (nitrogen is central atom)

E. $CH_3CH_2CH$

# Coordinate Covalent Bonds

**Ammonia**

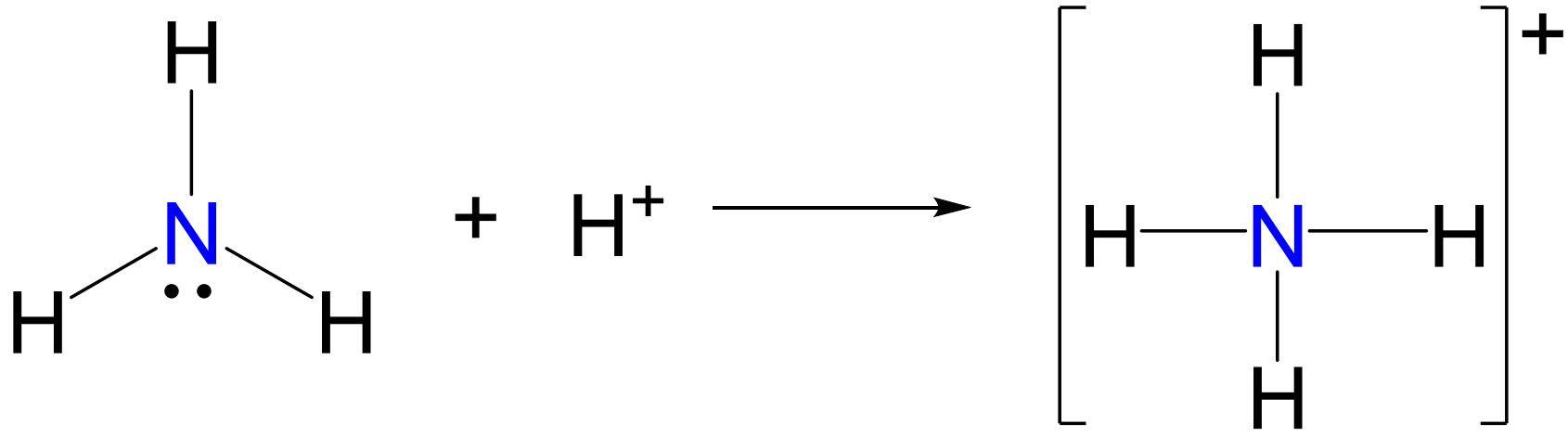
- Normal covalent bonds
- One electron from each atom shared between the two



# Coordinate Covalent Bonds

**Ammonium Ion**

- $H^+$ has no electrons
- N has lone pair
- Can still get $2e^-$s shared between them

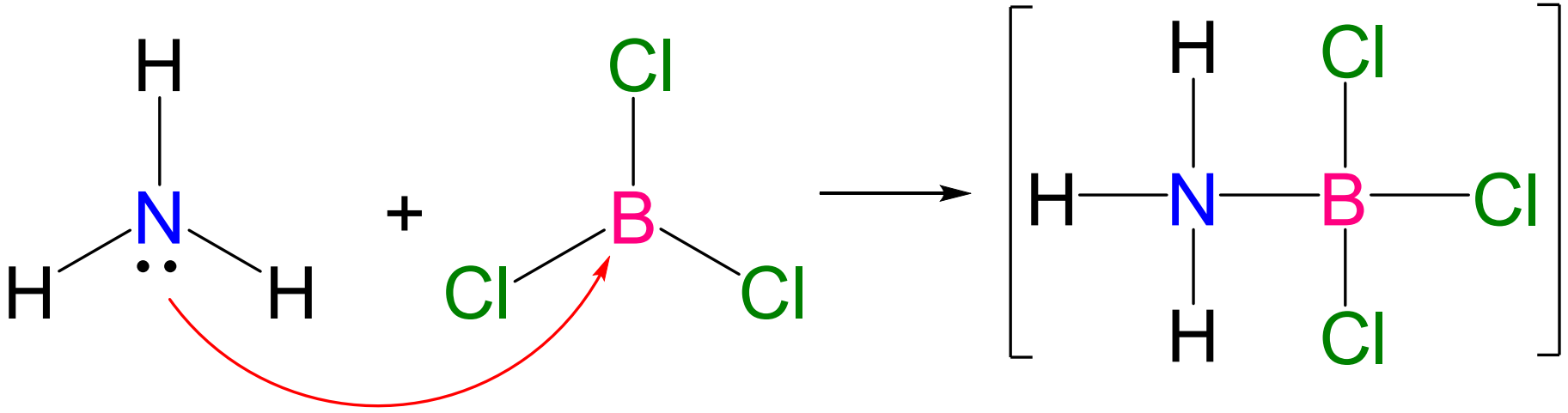


# Coordinate Covalent Bonds

- Both electrons of shared pair come from just one of two atoms
- Once bond formed, acts like any other covalent bond
- Can't tell where electrons came from after bond is formed
- Useful in understanding chemical reactions

# Coordinate Covalent Bonds

- Especially **boron** (electron deficient molecule) reacts with nitrogen compounds that contain lone pair of electrons



# Learning Check

**Ex.** What is the best Lewis Structure for **$HClO_4$**?

**$XeF_4$**

**$I_3^-$**

**$BrF_5$**